Search In Sorted Matrix

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The problem

- Rows and columns are sorted ascending; return [row, col] of target or [-1, -1].
- Start top-right and eliminate a row or column each step.

Intuition

- Start at the top-right: moving left decreases, moving down increases, so each step eliminates a row or column.

Approach

- 1. Start top-right
- 2. If too big, move left
- 3. If too small, move down
- 4. Return the coordinates or minus ones

Java solution

```java
public class SearchInSortedMatrix {
    public static int[] searchInSortedMatrix(int[][] m, int target) {
        int r = 0, c = m[0].length - 1;
        while (r < m.length && c >= 0) {
            if (m[r][c] == target) return new int[]{r, c};
            if (m[r][c] > target) c--; else r++;
        }
        return new int[]{-1, -1};
    }
}
```

Complexity

- Time: O(n + m)
- Space: O(1)

Recap

- Start at the top-right: moving left decreases, moving down increases, so each step eliminates a row or column.
- Complexity: O(n + m)